

Finals Lab Task 4. Design Power BI reports

Overview:

In this lab, you'll create a three-page report. You'll then publish it to the Power BI service, where you'll open and interact with the report.

In this lab, you learn how to:

- Design a report.
- Configure visual fields and format properties.
- Sync slicers.
- **Publish the report to the Power BI service.**
- Interact with a report and its visuals.
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This lab should take approximately 45 minutes.

Get started

To complete this exercise, first open a web browser and enter the following URL to download a zip file:

<https://drive.google.com/drive/folders/1TjDULUivAp6lpXe8DEWQJNp8rxcd9ilv?usp=sharing>

Extract the file to the C:\Users\Student\Downloads\04-design-report folder.

Open the **04-Starter-Sales Analysis.pbix** file.

Part 1. Design page 1

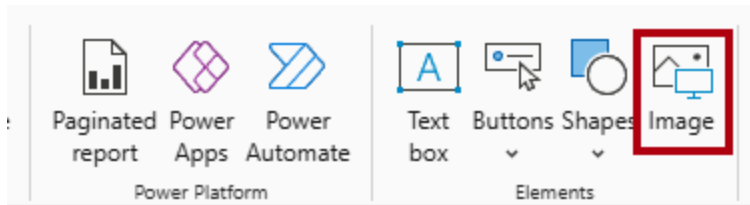
In this task, you'll design the first report page. When you've completed the design, the page will look like the following:



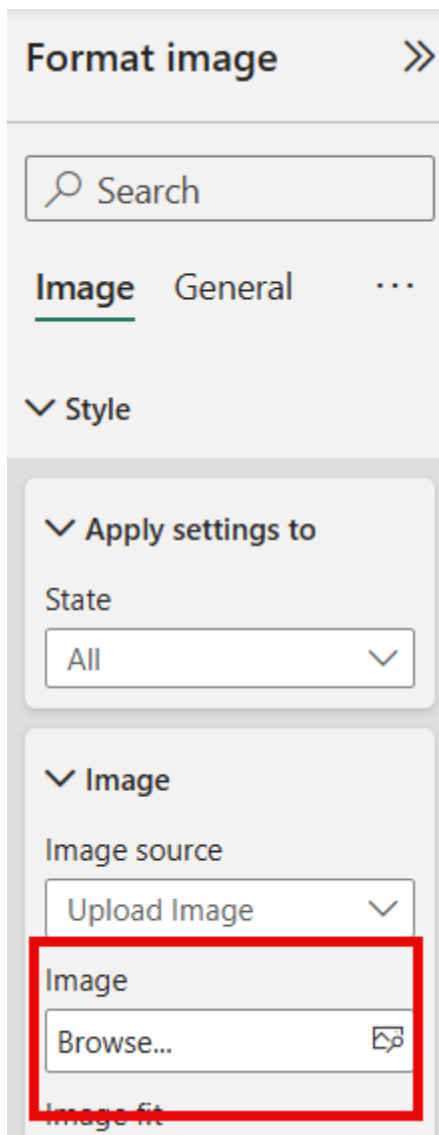
1. In Power BI Desktop, to rename the page, at the bottom-left, right-click **Page 1**, then select **Rename**. Rename the page as *Overview*.

Tip: You can also double-click the page name to rename it.

2. To add an image, on the **Insert** ribbon tab, from inside the **Elements** group, select **Image**.



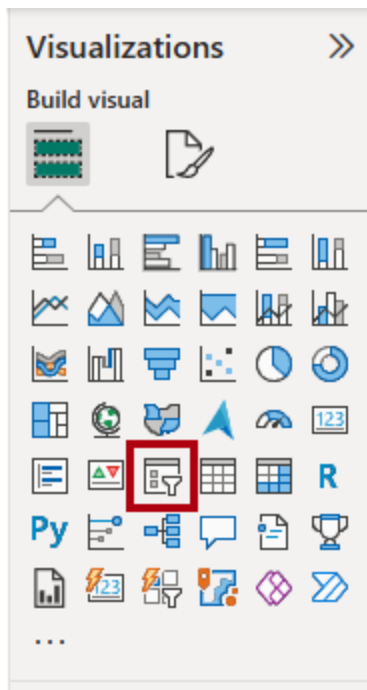
3. The **Format Image** pane will open beside the canvas. From this pane, select **Image > Browse** to select the image file in the next step.



4. In the **Open** window, navigate to the **C:\Users\Student\Downloads\08-design-report** folder.
5. Select the **AdventureWorksLogo.jpg** file, and then select **Open**.
6. Drag the image to position it at the top-left corner, and also drag the guide markers to resize it.

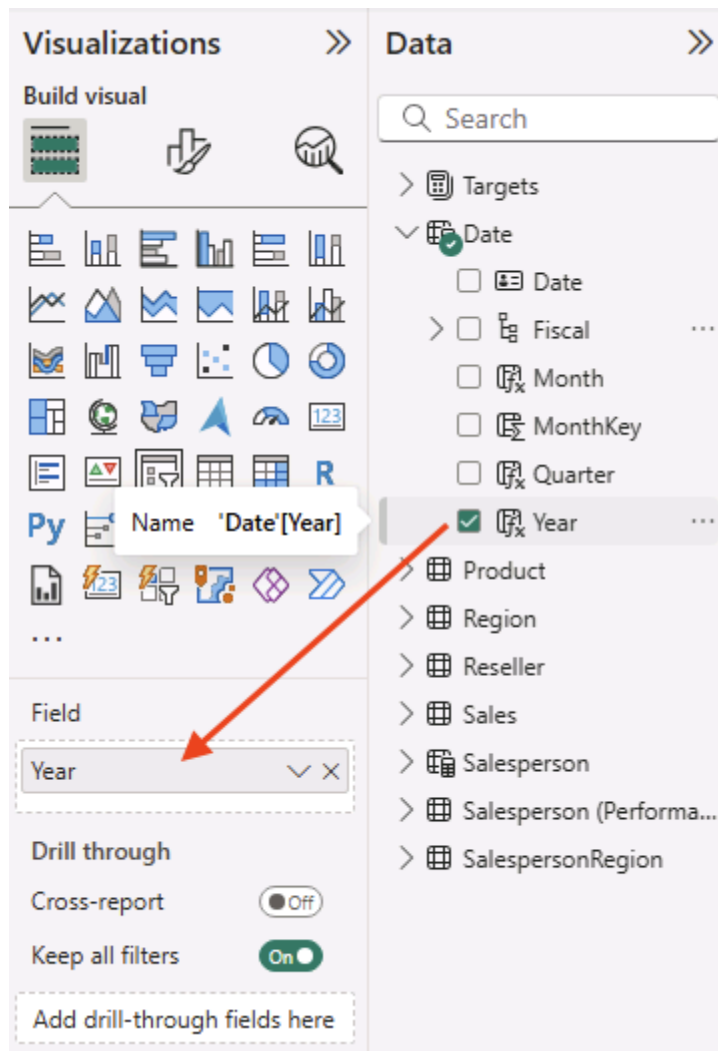


7. To add a slicer, first de-select the image by selecting an empty area of the report page, then select **Slicer** in the **Visualizations** pane.

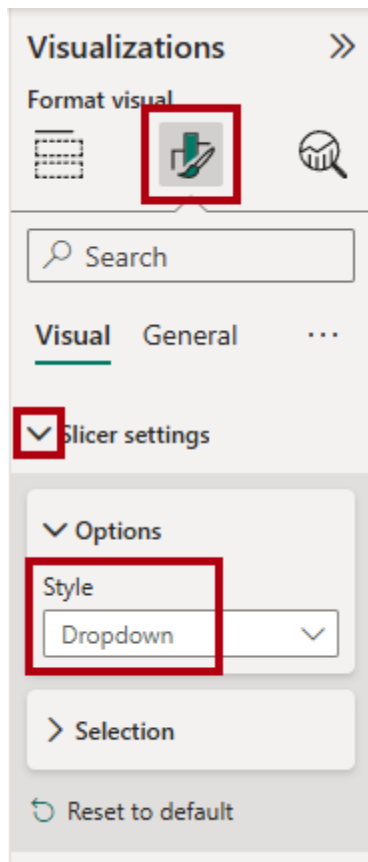


8. In the **Data** pane, drag the Date | Year field (not the Year level of the hierarchy) into the slicer **Field** well in the **Visualizations** pane.

The labs use a shorthand notation to reference a field. It will look like this: Date | Year. In this example, Date is the table name and Year is the field name.



9. To convert the slicer from a list to a dropdown, in the **Visualizations** pane, select **Format visual**. Expand **Slicer settings**, and then set the **Style** dropdown to **Dropdown**.



10. Resize and position the slicer so it sits beneath the image and is the same width as the image.



11. In the **Year** slicer, open the dropdown list, select **FY2020**, and then collapse the dropdown list.

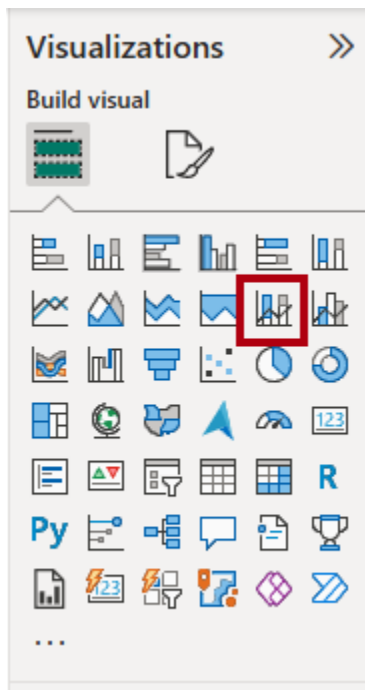
*The report page is now filtered by year **FY2020**.*



12. De-select the slicer by selecting an empty area of the report page.
13. Create a second slicer, based on the Region | Region field (not the Region level of the hierarchy).
14. Leave the slicer as a list, and then resize and position the slicer beneath the **Year** slicer.



15. De-select the slicer by selecting an empty area of the report page.
16. To add a chart to the page, in the **Visualizations** pane, select the **Line and Stacked Column Chart** visual type.



17. Resize and position the visual so it sits to the right of the logo, and so it fills the width of the report page.

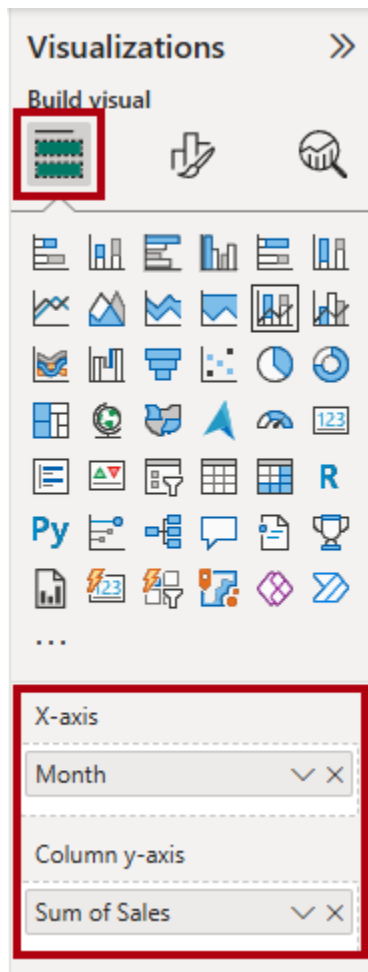


18. Drag and drop the following fields into the visual:

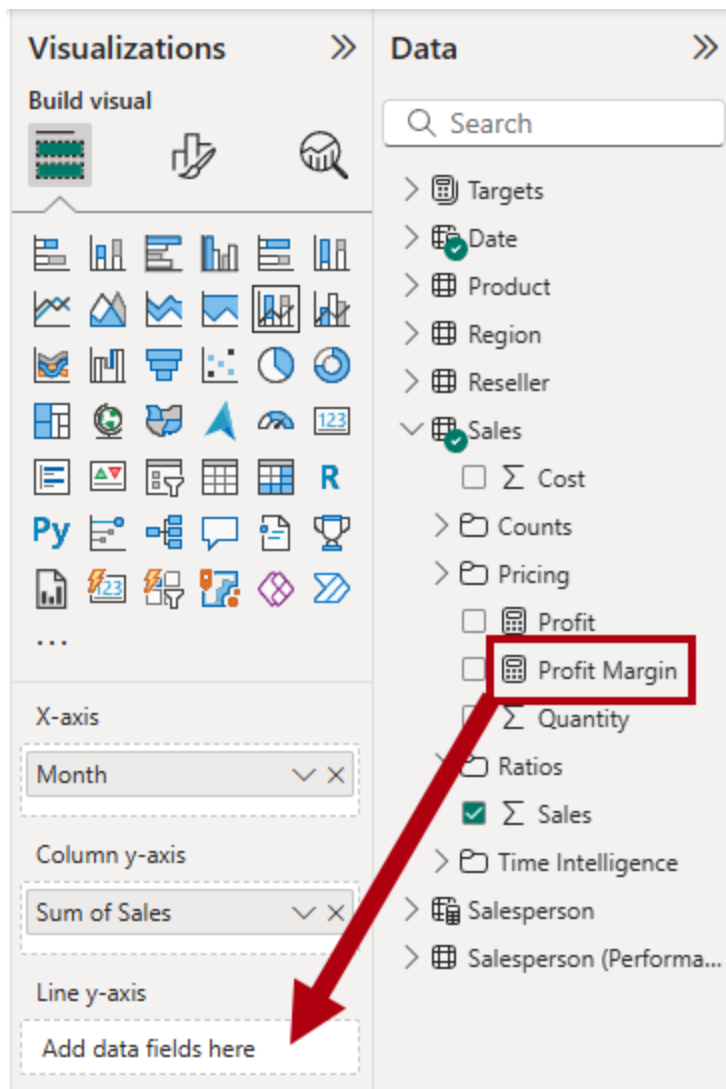
- Date | Month
- Sales | Sales

19. In the visual fields pane (located in the **Visualizations** pane), notice that the fields are assigned to the **X-axis** and **Column y-axis** wells.

By dragging fields into a visual, they're added to default wells. For precision, you can drag fields directly into the wells, as you'll do next.



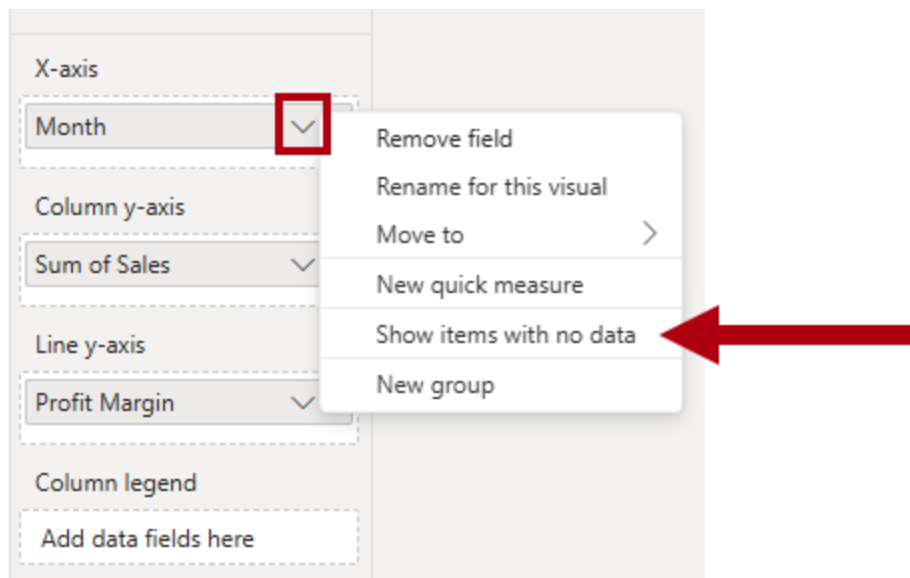
20. From the **Data** pane, drag the Sales | Profit Margin field into the **Line y-axis** well.



21. Notice that the visual only has 11 months.

The last month of the year, 2020 June, doesn't have any sales yet. By default, the visual has eliminated months with BLANK sales. You'll now configure the visual to show all months.

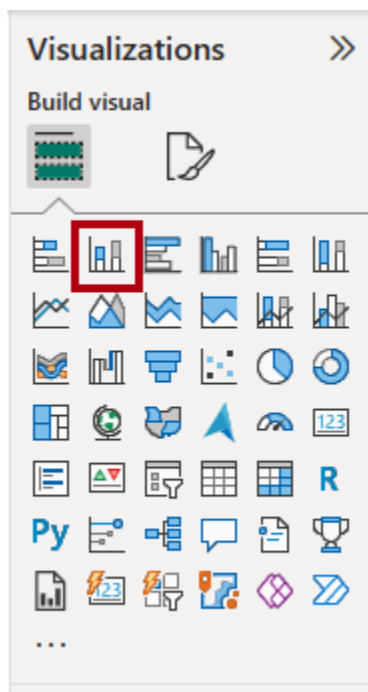
22. In the visual fields pane, in the **X-axis** well, for the **Month** field, select the down-arrow, and then select **Show items with no data**.



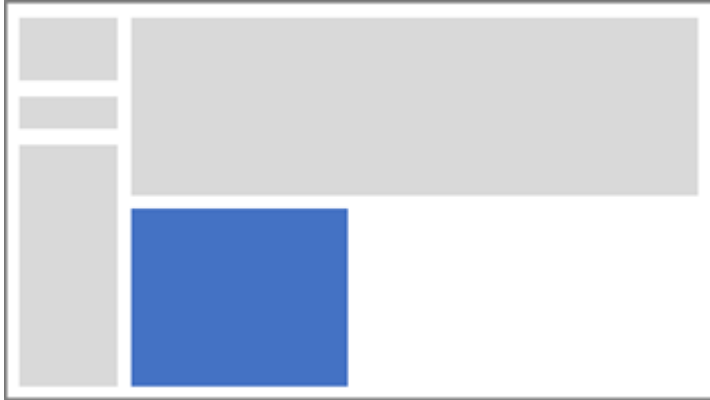
Notice that the month **2020 June** now appears.

23. De-select the chart by selecting an empty area of the report page.

24. To add a chart to the page, in the **Visualizations** pane, select the **Stacked Column Chart** visual type.



25. Resize and position the visual so it sits beneath the column/line chart, and so it fills half the width of the chart above.

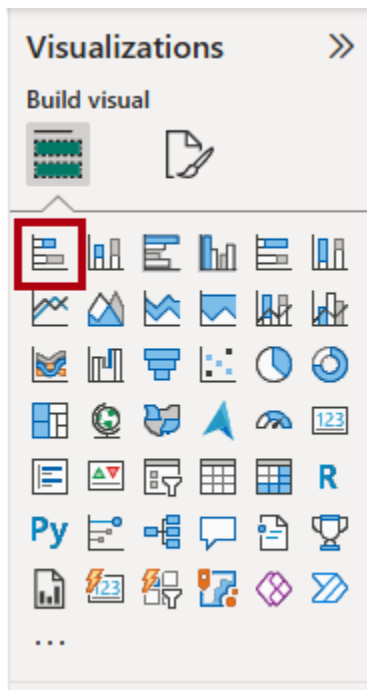


26. Add the following fields to the visual wells:

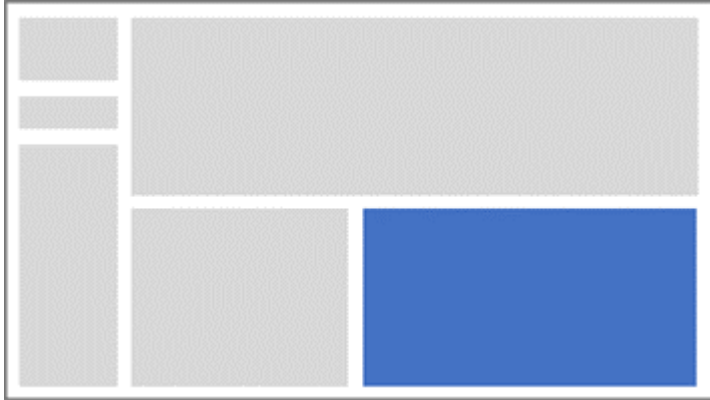
- X-axis: Region | Group
- Y-axis: Sales | Sales
- Legend: Product | Category

27. De-select the chart by selecting an empty area of the report page.

28. To add a chart to the page, in the **Visualizations** pane, select the **Stacked Bar Chart** visual type.



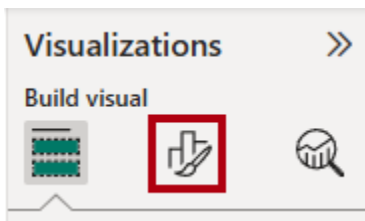
29. Resize and position the visual so it fills the remaining report page space.



30. Add the following fields to the visual wells:

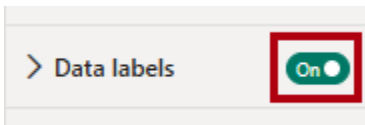
- Y-axis: Product | Category
- X-axis: Sales | Quantity

31. To format the visual, open the **Format** pane.



32. Expand the **Bars** section, and then in the **Color** group, set the **Color** property to a suitable color (to complement the column/line chart).

33. Set the **Data Labels** section to **On**.



34. Save the Power BI Desktop file.

The design of the first page is now complete

Part 2. Design Page 2

In this task, you'll design the second report page. When you've completed the design, the page will look like the following:

Region	Year	Orders	Sum of Sales	Sum of Cost	Profit	Profit Margin
☐ Australia	FY2018	739	\$16,429,043	\$16,297,680	\$131,362	0.80%
☐ Canada	FY2019	1255	\$27,979,780	\$26,768,232	\$1,211,548	4.33%
☐ Central	FY2020	1622	\$33,139,748	\$33,483,164	(\$343,416)	-1.04%
☐ France	Total	3616	\$77,548,570	\$76,549,076	\$999,495	1.29%
☐ Germany						
☐ Northeast						
☐ Northwest						
☐ Southeast						
☐ Southwest						
☐ United Kingdom						

Important: When detailed instructions have already been provided in the labs, the lab steps will provide more concise instructions. If you need the detailed instructions, you can refer back to other tasks in this lab.

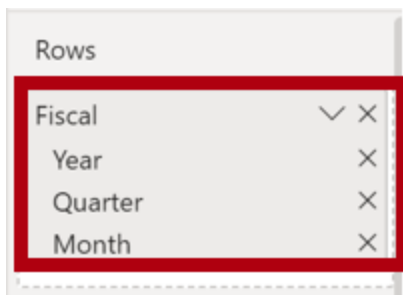
1. To create a new page, at the bottom-left, select the plus icon. When the page is added, rename it as *Profit*.
2. Add a slicer based on the Region | Region field.
3. Use the **Format** pane to show the *Select all* option (located in the **Slicer settings > Selection** section).
4. Resize and position the slicer so it sits at the left side of the report page, and so it's about half the page height.



5. Add a matrix visual, and resize and position it so it fills the remaining space of the report page



6. Add the Date | Fiscal hierarchy to the matrix **Rows** well.



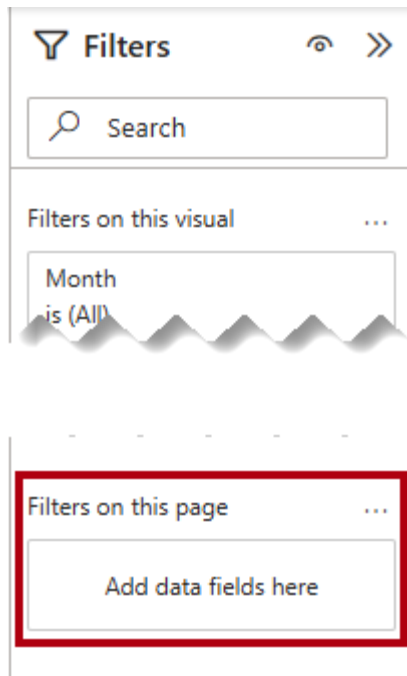
7. Add the following five Sales table fields to the **Values** well:

- Orders (from the Counts folder)
- Sales
- Cost
- Profit
- Profit Margin

The fields will show in the visual with a prefix showing the aggregation type; “Sum of Sales.”

Year	Orders	Sum of Sales	Sum of Cost	Profit	Profit Margin
⊕ FY2018	739	\$16,429,043	\$16,297,680	\$131,362	0.80%
⊕ FY2019	1255	\$27,979,780	\$26,768,232	\$1,211,548	4.33%
⊕ FY2020	1622	\$33,139,748	\$33,483,164	(\$343,416)	-1.04%
Total	3616	\$77,548,570	\$76,549,076	\$999,495	1.29%

8. In the **Filters** pane (located at the left of the **Visualizations** pane), notice the **Filters on this page** section (you may need to expand open the pane and scroll down).



9. From the **Data** pane, drag the Product | Category field into the **Filters on this page** well.

*Fields added to the **Filters** pane can achieve the same result as a slicer. One difference is they don't take up space on the report page. Another difference is that they can be configured to achieve more sophisticated filtering requirements.*

10. Inside the filter card, at the top right, select the arrow to collapse the card.

11. Add each of the following Product table fields to the **Filters on this page** well, collapsing each, directly beneath the Category field:

- Subcategory
- Product
- Color

Filters on this page ...

Category
is (All)

Subcategory
is (All)

Product
is (All)

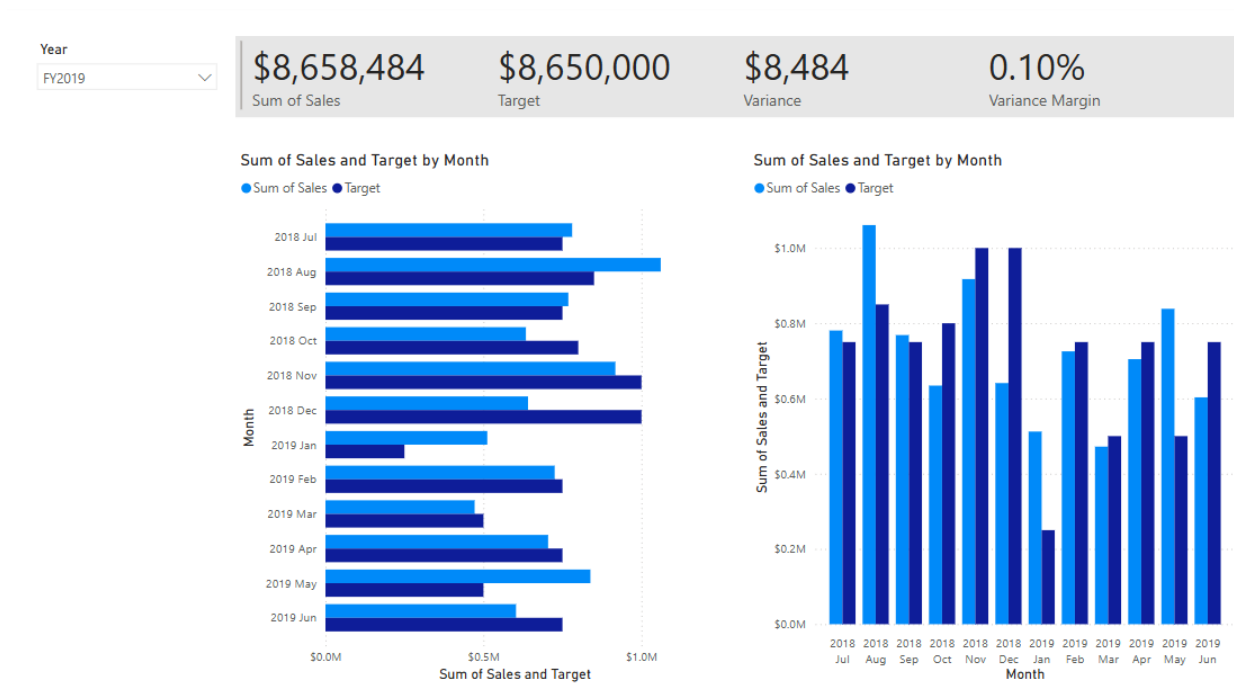
Color
is (All)

12. Save the Power BI Desktop file.

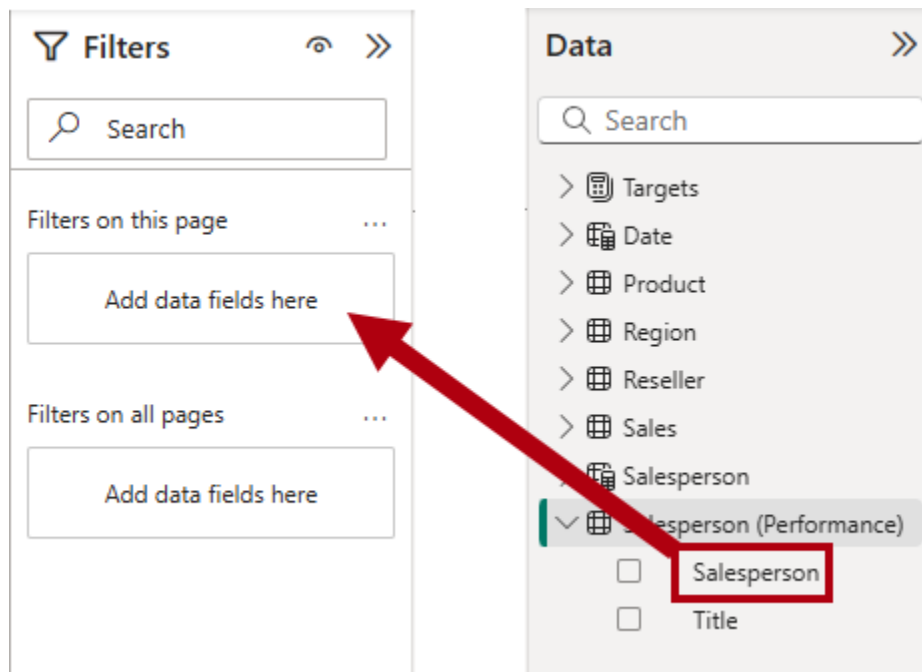
The design of the second page is now complete.

Part 3. Design page 3

In this task, you'll design the third—and final—report page. When you've completed the design, the page will look like the following:



1. Create a new page and rename it as *My Performance*.
2. To simulate the performance of row-level security filters, drag the Salesperson (Performance) | Salesperson field to the page level filters in the filter pane.



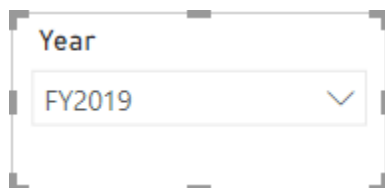
3. In the filter card, select **Michael Blythe**.

Data on the report page will now be filtered to display data for Michael Blythe only.

4. Add a dropdown slicer based on the Date | Year field, and then resize and position it so it sits at the top-left corner of the page.

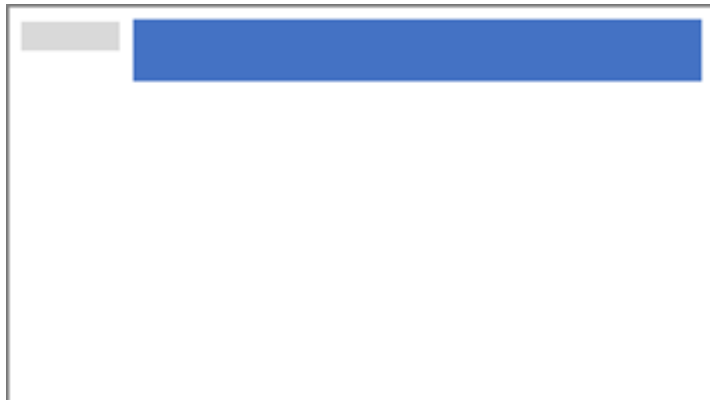
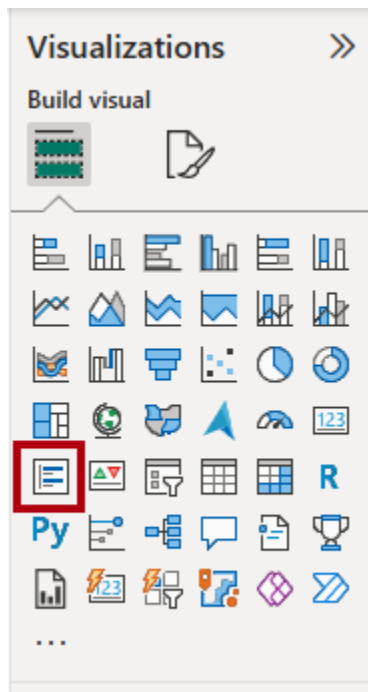


5. In the slicer, set the page to filter by **FY2019**.



6. Add a **Multi-row Card** visual to the page, and then resize and reposition it so it sits to the right of the slicer and fills the remaining width of the page.

If you don't see the **Multi-row Card** visual in the **Visualizations** pane, select the ellipsis (...) below the visuals and then **Restore default visuals**. You can now select **Multi-row Card**.



7. Add the following four fields to the visual:

- Sales | Sales
- Targets | Target
- Targets | Variance
- Targets | Variance Margin

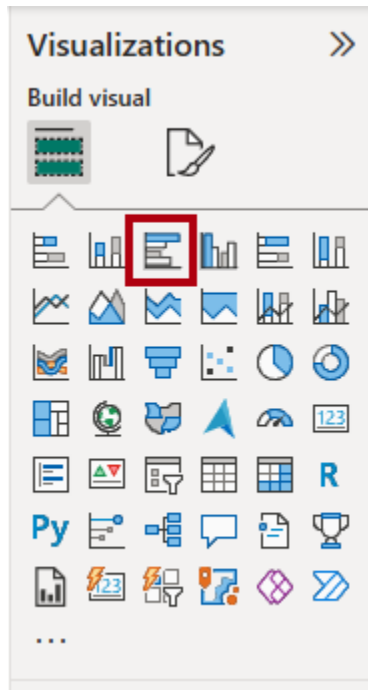
8. Format the visual:

- In the **Callout values** section, increase the font size property to **28pt**.

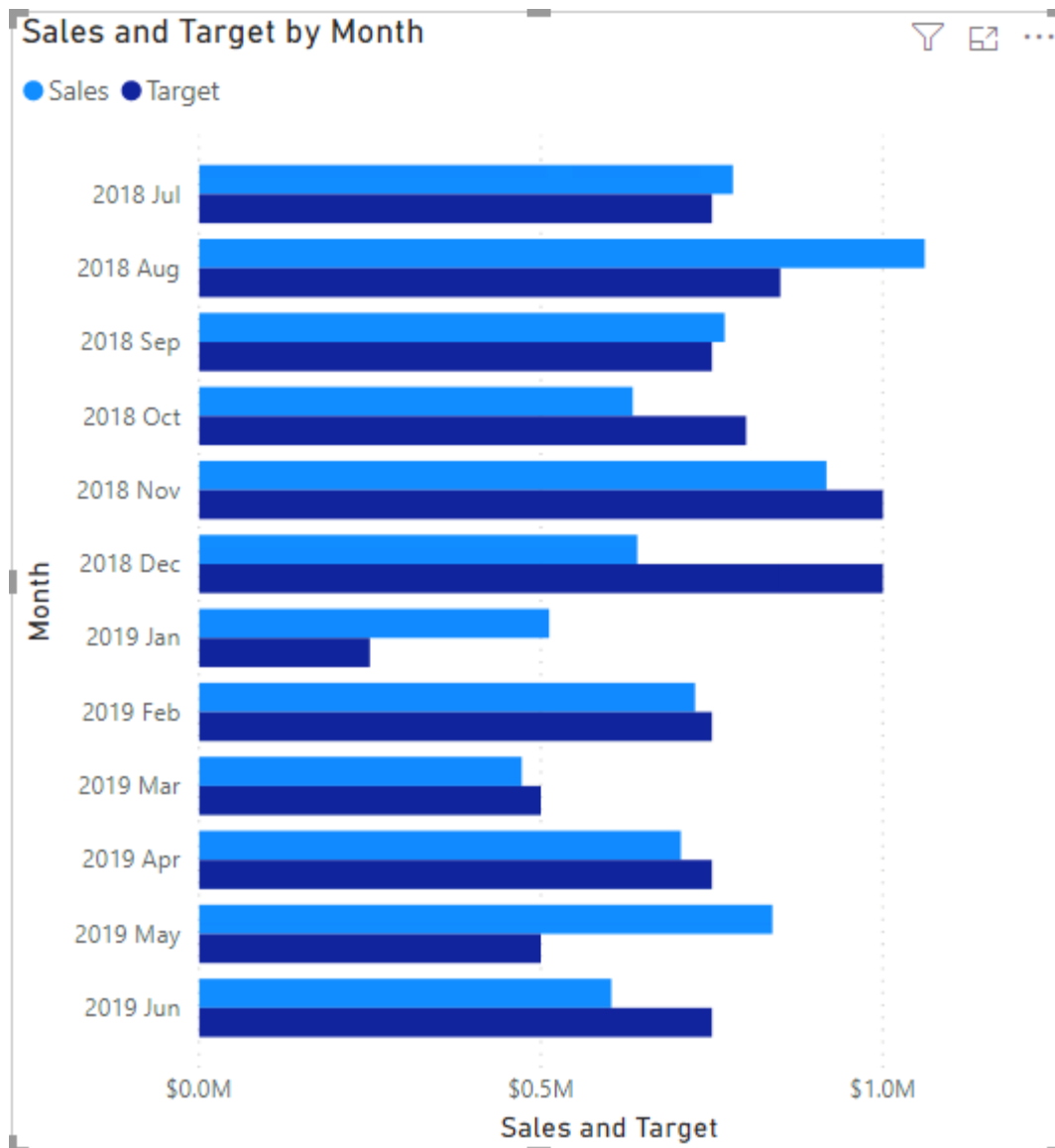
- In the **General** tab, in the **Effects** section, set the background color property to a light gray color (such as *White, 10% Darker*) to provide contrast.

\$8,658,484	\$8,650,000	\$8,484	0.10%
Sum of Sales	Target	Variance	Variance Margin

9. Add a **Clustered Bar Chart** visual to the page, and then resize and position it so it sits beneath the multi-row card visual and fills the remaining height of the page, and half the width of the multi-row card visual.



10. Add the following fields to the visual wells:
 - Y-axis: Date | Month
 - X-axis: Sales | Sales and Targets | Target

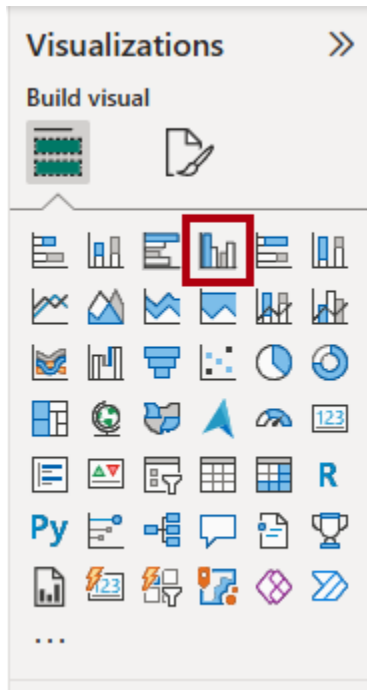


11. To create a copy of the visual, press **Ctrl+C**, and then press **Ctrl+V**.

12. Position the new visual to the right of the original visual.



13. To modify the visualization type, in the **Visualizations** pane, select **Clustered Column Chart**.



It's now possible to see the same data expressed by two different visualization types. The design of the final page is complete.

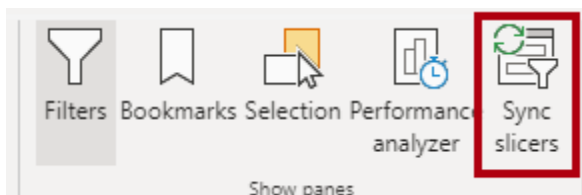
Sync slicers

In this task, you'll sync the *Year* and *Region* slicers.

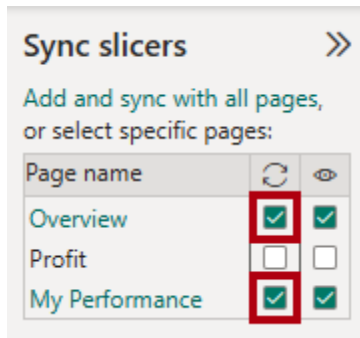
1. On the *Overview* page, set the *Year* slicer to **FY2018**.
2. Go to the *My Performance* page, and then notice that the *Year* slicer has a different value.

When slicers aren't synced, it can contribute to misrepresentation of data and frustration for report users. You'll now sync the report slicers.

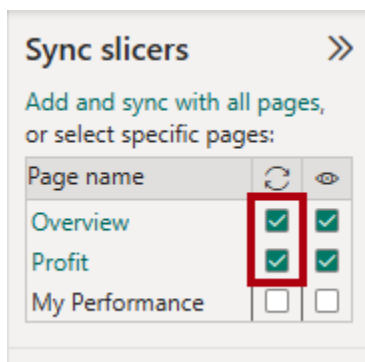
3. Return to the *Overview* page, and then select the *Year* slicer.
4. On the **View** ribbon tab, from inside the **Show Panes** group, select **Sync Slicers**.



5. In the **Sync Slicers** pane (at the left of the **Visualizations** pane), in the second column (which represents syncing), check the checkboxes for the *Overview* and *My Performance* pages.



6. On the *Overview* page, select the *Region* slicer.
7. Sync the slicer with the *Overview* and *Profit* pages.



8. Test the sync slicers by selecting different filter options, and then verifying that the synced slicers filter by the same selection.
9. To close the **Sync Slicer** pane, select the **Sync Slicer** option located on the **View** ribbon tab.
- 10. Save File as 04-Sales_Analysis_complete.pbix**
11. Upload the **pbix file** and screenshot of your output and submit in gclass and post in your portfolio (with Task description and screenshot)